Part A: First Normal Form (1NF)

Step 1: Definition

A relation is in first normal form if the domain of every attribute is atomic, where each attribute holds a single, indivisible value rather than a set or a composite of several values. Each relation must also have a primary key, which was added in Part 1.

Step 2: Check Relations

Relation	Attributes	Atomic?	Explanation
Employee	empID, emp_fname, emp_initials, emp_lname, job, deptID, salary	Yes	Name is already split into first, initials, and last name.
Project	projID, title, phase, budget, funds	Yes	All attributes contain indivisible values.
Assigned	empID, projID, roleID, role_description, role_level, start_date, end_date, assigned_status	Yes	Dates ignored per Note 3; all values are atomic.
Department	deptID, deptName, **location**	**No**	`location` is a composite value containing address details.

Step 3: Why is Location non-atomic

`Department.location` packs five separate components into a single string. For example, the value `'200 University Av W, Waterloo, ON, N2L3G1'` decomposes as:

Component	Value
streetNum	200
streetName	University Av W
cityName	Waterloo
province	ON
postalcode	N2L3G1

Because one attribute value contains several independent components, a single component (for example, every department in Toronto) cannot be processed without parsing. Therefore, the schema from Part 1 is not in 1NF.

Step 4: Replace (No new tables)

Each department has exactly one location, so there is no repeating group. The components can be stored directly in Department, and no new table is needed (Note 1). Thus, we can replace `location` with five atomic attributes:

Department(*deptID*, *deptName*, *streetNum*, *streetName*, *cityName*, *province*, *postalcode*)

Step 5: Functional Dependencies

The Department dependency becomes:

$$deptID \rightarrow deptName,\ streetNum,\ streetName,\ cityName,\ province,\ postalcode$$

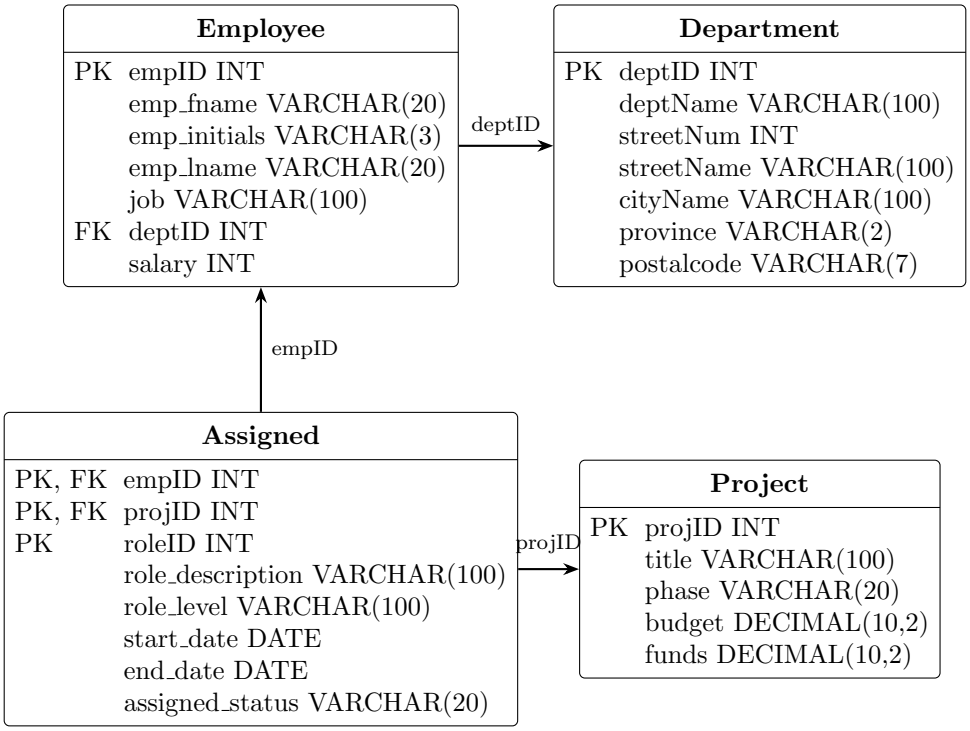
All other functional dependencies from Table 1 are unchanged. Postal codes are ignored (Note 3).

Step 6: Resulting 1NF schema

Primary keys are underlined; foreign keys are marked FK.

- Department(*deptID*, *deptName*, *streetNum*, *streetName*, *cityName*, *province*, *postalcode*)
- Employee(*empID*, *emp_fname*, *emp_initials*, *emp_lname*, *job*, *deptID* $^{\text{FK}\rightarrow\text{Department}}$, *salary*)
- Project(*projID*, *title*, *phase*, *budget*, *funds*)
- Assigned(*empID* $^{\text{FK}\rightarrow\text{Employee}}$, *projID* $^{\text{FK}\rightarrow\text{Project}}$, *roleID*, *role_description*, *role_level*, *start_date*, *end_date*, *ass*

Step 7: 1NF relational schema diagram



Arrows point from a foreign key to the primary key it references.

Part B: Boyce–Codd Normal Form (BCNF)

Step 1: Definition

A relation schema R is in BCNF with respect to a set F of functional dependencies if, for every non-trivial dependency $\alpha \to \beta$ in F^+ where $\alpha, \beta \subseteq R$, the determinant α is a superkey for R. Formally, for each dependency $\alpha \to \beta$, at least one of the following holds:

- $\alpha \to \beta$ is trivial ($\beta \subseteq \alpha$), or
- α is a superkey for R.

Step 2: Test and the dependency set

To test for BCNF, we evaluate the attribute set closures of subsets of R with respect to F. If a non-trivial dependency $\alpha \to \beta$ exists where α is not a superkey, the relation violates BCNF. Closures are restricted to the relation being evaluated by intersecting with R, i.e., $(\alpha)^+ \cap R$.

The full set of functional dependencies F (with the updated Department dependency from Part A) is:

$f_1 : \textit{deptID} \to \textit{deptName},\ \textit{streetNum},\ \textit{streetName},\ \textit{cityName},\ \textit{province},\ \textit{postalcode}$

$f_2 : \textit{projID} \to \textit{title},\ \textit{phase},\ \textit{budget},\ \textit{funds}$

$f_3 : \textit{empID} \to \textit{emp_fname},\ \textit{emp_initials},\ \textit{emp_lname},\ \textit{job},\ \textit{deptID},\ \textit{salary}$

$f_4 : \textit{empID},\ \textit{projID},\ \textit{roleID} \to \textit{role_description},\ \textit{role_level},\ \textit{start_date},\ \textit{end_date},\ \textit{assigned_status}$

$f_5 : \textit{roleID} \to \textit{role_description},\ \textit{role_level}$

The closure algorithm only expands attributes through dependencies whose left-hand sides are present in the current attribute set. The determinants in F are $\{\textit{deptID}\}$, $\{\textit{projID}\}$, $\{\textit{empID}\}$, $\{\textit{roleID}\}$, and $\{\textit{empID}, \textit{projID}, \textit{roleID}\}$. Any subset containing none of these determinants yields a trivial closure $(\alpha)^+ = \alpha$, which cannot violate BCNF. Thus, we only need to test subsets formed from these key attributes.

Step 3: Test Department

$R = \text{Department}(\textit{deptID},\ \textit{deptName},\ \textit{streetNum},\ \textit{streetName},\ \textit{cityName},\ \textit{province},\ \textit{postalcode})$

$(\textit{deptID})^+ \cap R = \{\textit{deptID},\ \textit{deptName},\ \textit{streetNum},\ \textit{streetName},\ \textit{cityName},\ \textit{province},\ \textit{postalcode}\} = R$

Every subset containing *deptID* is a superkey, and subsets without *deptID* produce trivial closures. Therefore, **Department is in BCNF**.

Step 4: Test Project

$R = \text{Project}(\textit{projID},\ \textit{title},\ \textit{phase},\ \textit{budget},\ \textit{funds})$

$$(\textit{projID})^+ \cap R = \{\textit{projID},\ \textit{title},\ \textit{phase},\ \textit{budget},\ \textit{funds}\} = R$$

Every subset containing *projID* is a superkey, and all other subsets yield only trivial dependencies. Therefore, **Project is in BCNF**.

Step 5: Test Employee

$R = \text{Employee}(\textit{empID},\ \textit{emp_fname},\ \textit{emp_initials},\ \textit{emp_lname},\ \textit{job},\ \textit{deptID},\ \textit{salary})$

$$(\textit{empID})^+ \cap R = \{\textit{empID},\ \textit{emp_fname},\ \textit{emp_initials},\ \textit{emp_lname},\ \textit{job},\ \textit{deptID},\ \textit{salary}\} = R$$
$$(\textit{deptID})^+ \cap R = \{\textit{deptID}\}$$

While $(\textit{deptID})^+$ determines Department attributes, none except *deptID* exist in Employee. Within Employee, *deptID* only determines itself. Since every subset containing *empID* is a superkey, **Employee is in BCNF**.

Step 6: Test Assigned

$R = \text{Assigned}(\textit{empID}, \textit{projID}, \textit{roleID}, \textit{role_description}, \textit{role_level}, \textit{start_date}, \textit{end_date}, \textit{assigned_status})$

Let $E = \textit{empID}$, $P = \textit{projID}$, $I = \textit{roleID}$, $D = \textit{role_description}$, $L = \textit{role_level}$, $S = \textit{start_date}$, $N = \textit{end_date}$, and $T = \textit{assigned_status}$, so $R = EPIDLSNT$. Within R, the relevant dependencies are $f_4 : EPI \to DLSNT$ and $f_5 : I \to DL$.

Closures of the subsets of $\{E, P, I\}$, restricted to R:

$$(E)^+ = E \qquad (P)^+ = P \qquad (I)^+ = IDL$$
$$(EP)^+ = EP \qquad (EI)^+ = EIDL \qquad (PI)^+ = PIDL$$
$$(EPI)^+ = EPIDLSNT = R$$

Adding any of D, L, S, N, T to one of these subsets only adds those attributes to its closure, and any subset of $\{D, L, S, N, T\}$ alone has $(\alpha)^+ = \alpha$. So the only superkeys are the subsets containing EPI.

- The closure $(I)^+ = IDL$ shows that $I \to DL$ (*roleID* $\to$ *role_description*, *role_level*), which is non-trivial since $D, L \notin \{I\}$.
- Since I is not a superkey for Assigned ($E, P, S, N, T \notin (I)^+$), it follows that $I \to DL$ violates BCNF.
- Conclusion: **Assigned is not in BCNF**.

($EI \to DL$ and $PI \to DL$ also violate BCNF, but both follow from $I \to DL$ by augmentation, so removing $I \to DL$ removes them as well.)

This causes redundancy: a role's description and level are repeated in every assignment with that role (e.g. roleID 3, Manager, L4, is stored four times).

Step 7: Decompose Assigned (BCNF decomposition procedure)

Iteration 1. *result* := {Department, Employee, Project, Assigned}. Assigned is not in BCNF. Let $\alpha = \{\textit{roleID}\}$ and $\beta = \{\textit{role_description},\ \textit{role_level}\}$. Then $\alpha \to \beta$ is non-trivial, holds on Assigned, $\alpha \cap \beta = \emptyset$, and α is not a superkey for Assigned.

- Remove Assigned from *result*.
- Add Assigned $- \beta =$ (*empID*, *projID*, *roleID*, *start_date*, *end_date*, *assigned_status*), which keeps the name **Assigned**.

- Add $\alpha\beta = (\textit{roleID},\ \textit{role_description},\ \textit{role_level})$, named **Role**.

Iteration 2. Every relation in *result* is in BCNF (Steps 3–5 and Step 8), so *done* := true.

Lossless join. The BCNF decomposition procedure always produces a lossless-join decomposition. Directly: Role $\cap$ Assigned = $\{\textit{roleID}\}$ and $\textit{roleID} \rightarrow$ Role, so the common attribute is a superkey of Role.

Dependency preservation. Let F_i be the dependencies in F^+ that include only attributes of each relation. F_{Role} contains $I \rightarrow DL$ and F_{Assigned} contains $EPI \rightarrow SNT$. From $EPI \rightarrow I$ (trivial) and $I \rightarrow DL$ we get $EPI \rightarrow DL$ by transitivity, and with $EPI \rightarrow SNT$ the union rule gives $EPI \rightarrow DLSNT$, which is f_4. f_1, f_2, f_3 are unchanged. Therefore $(F_1 \cup \cdots \cup F_n)^+ = F^+$, and the decomposition is dependency preserving.

Step 8: Test the new relations (general BCNF test)

Role $= (I, D, L)$:

$$(I)^+ = IDL \qquad (D)^+ = D \qquad (L)^+ = L \qquad (DL)^+ = DL$$

Since there are only trivial dependencies (among attributes of Role) and $I \rightarrow DL$ (where I is a superkey for Role), it follows that **Role is in BCNF**.

Assigned $= (E, P, I, S, N, T)$, closures restricted to the new relation:

$$(E)^+ = E \qquad (P)^+ = P \qquad (I)^+ = I$$
$$(EP)^+ = EP \qquad (EI)^+ = EI \qquad (PI)^+ = PI$$
$$(EPI)^+ = EPISNT = \text{Assigned}$$

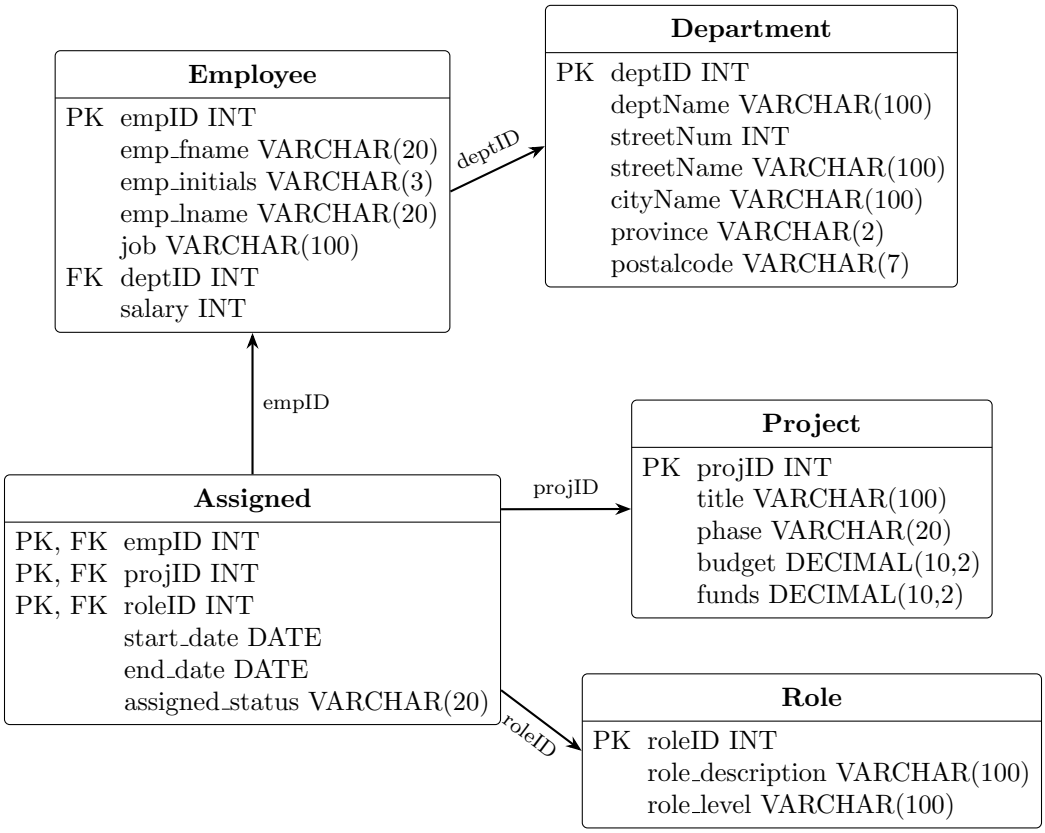
D and L are no longer in this relation, so I only determines itself here. Since there are only trivial dependencies (among attributes of Assigned) and $EPI \rightarrow SNT$ (where EPI is a superkey for Assigned), it follows that **Assigned is in BCNF**.

Step 9: Resulting BCNF schema

The schema has five tables; Role is the only new one. Primary keys are underlined; foreign keys are marked FK.

- Department(<u>*deptID*</u>, *deptName*, *streetNum*, *streetName*, *cityName*, *province*, *postalcode*)
- Employee(<u>*empID*</u>, *emp_fname*, *emp_initials*, *emp_lname*, *job*, *deptID* $^{\text{FK}\rightarrow\text{Department}}$, *salary*)
- Project(<u>*projID*</u>, *title*, *phase*, *budget*, *funds*)
- Role(<u>*roleID*</u>, *role_description*, *role_level*)
- Assigned(<u>*empID*</u> $^{\text{FK}\rightarrow\text{Employee}}$, <u>*projID*</u> $^{\text{FK}\rightarrow\text{Project}}$, <u>*roleID*</u> $^{\text{FK}\rightarrow\text{Role}}$, *start_date*, *end_date*, *assigned_status*)

Step 10: BCNF relational schema diagram



Arrows point from a foreign key to the primary key it references.